

Primavera P6 Baseline Risk Intelligence Engine

Identifying high-risk activities and
vulnerable work packages before project
execution

Python | Primavera P6 | PMO Analytics

The Problem

Activity ID	Actual Finish	BL Project Finish	Actual Duration	Predecessors	BL Project Duration	Successors	Activity Status	Critical
HBTF-2 HBTF - Package 2 (Main Contract Works)		23-Jul-14	0		641			
HBTF-2.1 Project Milestone		23-Jul-14	0		641			
HBTF-2.2 Mobilization		05-Jun-14	0		599			
HBTF-2.3 Engineering		02-Oct-13	0		399			
HBTF-2.3.4 Required Data		30-Jun-12	0		21			
HCERDA10		30-Jun-12	0	H0000	21	HEMSSSS0020	Not Startec	☐
HCERDA000		30-Jun-12	0	H0000	21	HCSWB4Z1010	Not Startec	☐
HBTF-2.3.3 Subcontractors		08-Aug-12	0		55			
HBTF-2.3.3.1 Nomination		23-Jul-12	0		41			
HCESCN030		15-Jul-12	0	H0000, HMMAZ030	20	HCESCA030	Not Startec	☐
HCESCN020		16-Jun-12	0	H0000	10	HCESCA020	Not Startec	☐
HCESCN010		16-Jun-12	0	H0000	10	HCESCA010	Not Startec	☒
HCESCN000		23-Jul-12	0	H0000, HMMAZ030	29	HCESCA000, HEMSSME	Not Startec	☒
HBTF-2.3.3.2 Approval		08-Aug-12	0		45			
HCESCA030		31-Jul-12	0	HCESCN030	14	HEPSCVS0010	Not Startec	☐
HCESCA020		03-Jul-12	0	HCESCN020	14	HEMSSSAL0000, HEMSS	Not Startec	☐
HCESCA010		03-Jul-12	0	HCESCN010	14	HEMSSSS0000, HEMSS	Not Startec	☒
HCESCA000		08-Aug-12	0	HCESCN000	14	HEMSSME1050, HEMS	Not Startec	☒
HBTF-2.3.1 Shopdrawings		02-Oct-13	0		399			
HBTF-2.3.1.1 Submission		16-Sep-13	0		385			
HBTF-2.3.1.1.1 Civil		10-Nov-12	0		127			
HESDSCI0010		10-Nov-12	0	HESDSCI0000	125	HESDACI0010, HESDAI	Not Startec	☐
HESDSCI0000		06-Jun-12	0	H0000	2	HESDACI0000, HESDSI	Not Startec	☐
HBTF-2.3.1.1.2 Finishing		05-May-13	0		150			
HCSDSFw040		05-May-13	0	HCSDSFw020, HESDSSI	150	HCSDAFw040, HCSDAI	Not Startec	☐
HCSDSFw030		05-May-13	0	HESDSSS0000	150	HCSDAFw030, HEMSS	Not Startec	☐
HCSDSFw020		05-May-13	0	HESDSSS0000	150	HCSDAFw020, HEMSS	Not Startec	☐
HCSDSFw010		05-May-13	0	HESDSSS0000	150	HCSDAFw010, HEMSS	Not Startec	☐
HBTF-2.3.1.1.4 MEP All Works		16-Sep-13	0		330			
HCSNMEB020		16-Sep-13	0	HCSNMEB000	220	HCSNMEB020, HCSNMEB000	Not Startec	☐

Large Primavera schedules can contain thousands of activities, multiple work packages, approvals, subcontractors, procurement, engineering, and construction sequences. Traditional schedule reviews identify critical path activities, but they do not explain where management should focus first.

Package Exposure Logic

```
package_exposure["baseline_exposure_score"] = (  
    package_exposure["critical_density"] * 0.40  
    + package_exposure["critical_activities"] * 0.30  
    + package_exposure["total_successors"] * 0.20  
    + package_exposure["total_remaining_days"] * 0.01  
)  
  
package_exposure.sort_values(  
    "baseline_exposure_score",  
    ascending=False  
) .head(20)
```

Package Exposure Analysis

wbs_name	total_activities	baseline_exposure_score
Electrical Works	567	257.824215
Wall Finishes	444	247.837072
Floor Finishes	423	178.606690
Ceiling Finishes	301	171.473198
Air Conditioning	260	112.030000
Steel Structure Works	144	103.612222
Retaining Walls	192	96.780000
Mecellanuos	183	89.120000
Drainage	156	83.600000
Miscellaneous	140	77.270000
MEP Materials	46	69.417391

Top Risk Activities

The engine reduced 4,000+ activities to a shortlist of executive-level risks.

	task_code	task_name	wbs_name	activity_risk_score	risk_category	risk_reason
782	HCESCA000	s/c approval: MEP subcontractor	Approval	84.5	Critical	Low/zero float + High successor impact + Locat...
785	HESDSS0000	SD submission: Steel Strucutre shopdrawings	Steel Strucuture	79.5	High	Low/zero float + Long duration exposure + Loca...
786	HESDASS0000	SD approval: Steel Strucutre shopdrawings	Steel Strucuture	77.5	High	Low/zero float + Long duration exposure + Loca...
855	HEPSAME1160	Diffusers	MEP Materials	74.5	High	Low/zero float + Long duration exposure
858	HEPSAME1300	CableTrays	MEP Materials	74.5	High	Low/zero float + Long duration exposure
3706	HCSWAF000	Construct of left over areas (crane # 3)	Casting and Finishing of Left over slabs	72.0	High	Low/zero float + Long duration exposure + Loca...

Results & Insights

- 4218 Activities Analyzed
- 197 Critical Activities Identified
- 9 Executive-Level Risk Activities
- Steel Structure Works Critical Density $\approx 49\%$
- MEP Approval Identified as Major Schedule Driver
- Next Phase: Interactive PMO Dashboard